

NOTICE INVITING TENDER (NIT)

SOLAR ROOFTOP INSTALLATION AND O&M

Tender Reference No.: REDC/SOLAR/2025-26/041

Date of Issue: 20 January 2026

RAJASTHAN ENERGY DEVELOPMENT CORPORATION LTD (REDC)

Renewable Energy Division

Urja Bhavan, Jyoti Nagar, Jaipur – 302005

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SECTION 1: GENERAL INFORMATION

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1.1 NAME OF WORK

Design, Supply, Installation, Testing, Commissioning and 7-Year Comprehensive Operation & Maintenance of Grid-Connected Rooftop Solar Photovoltaic (SPV) Systems aggregating 5 MWp capacity across 120 Government Buildings in Jaipur, Jodhpur, and Udaipur districts under PM Surya Ghar Yojana Phase-II.

1.2 ESTIMATED PROJECT COST

Rs. 28,75,00,000/- (Rupees Twenty-Eight Crore Seventy-Five Lakhs Only)

1.3 EARNEST MONEY DEPOSIT (EMD)

Rs. 57,50,000/- (Rupees Fifty-Seven Lakhs Fifty Thousand Only)

EMD via Bank Guarantee / DD in favour of REDC, Jaipur.

Valid for 180 days from bid submission date.

1.4 PERFORMANCE SECURITY

5% of Contract Value as Bank Guarantee, valid for contract period + 90 days.

1.5 DEFECT LIABILITY PERIOD

12 months from date of Commissioning Certificate for each site.

1.6 TENDER DOCUMENT FEE

Rs. 15,000/- (Non-refundable) via RTGS to A/C No. 9876543210, IFSC: PUNB0012345

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SECTION 2: SCHEDULE OF KEY DATES

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- 2.1 DATE OF PUBLICATION : 20 January 2026
- 2.2 SITE VISIT (MANDATORY) : 28-29 January 2026
- 2.3 PRE-BID CONFERENCE : 01 February 2026 at 10:30 AM
- 2.4 LAST DATE FOR SUBMISSION OF QUERIES : 05 February 2026
- 2.5 RESPONSE TO QUERIES : 10 February 2026
- 2.6 LAST DATE FOR BID SUBMISSION : 25 February 2026 up to 4:00 PM
- 2.7 BID OPENING DATE (TECHNICAL) : 26 February 2026 at 11:00 AM
- 2.8 BID OPENING DATE (FINANCIAL) : To be notified after Technical Evaluation
- 2.9 WORK COMPLETION PERIOD : 9 months from date of work order
- 2.10 O&M PERIOD : 7 years post-commissioning

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SECTION 3: ELIGIBILITY CRITERIA

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3.1 LEGAL ENTITY

Bidder must be a company/firm/LLP with valid GST Registration, PAN, and MSME/Udyam certificate (if applicable). Must be empanelled or eligible for empanelment with MNRE / RRECL.

3.2 FINANCIAL ELIGIBILITY

- 3.2.1 Average Annual Turnover: Minimum Rs. 20 Crores per annum for last 3 financial years (FY 2022-23, 2023-24, 2024-25).
- 3.2.2 Net Worth: Positive net worth as per latest audited financials.
- 3.2.3 Credit Facility: Minimum Rs. 5 Crores (bank solvency certificate required).

3.3 TECHNICAL ELIGIBILITY – SIMILAR WORK

Bidder must have completed at least ONE rooftop/ground-mounted solar project of not less than 1 MWp capacity OR TWO projects each of not less than 500 KWp for any Government / PSU / Municipal / Industrial client in the last 5 years.

3.4 CHANNEL PARTNER / EMPANELMENT

Bidder shall be an authorised channel partner or system integrator of a MNRE-approved solar panel manufacturer and MNRE-approved inverter supplier.

3.5 KEY PERSONNEL

- 3.5.1 Project Manager: B.E./B.Tech (Electrical/Electronics), 8 years experience

in solar/renewable energy projects.

3.5.2 Electrical Engineer (per district): B.E. Electrical with 4 years experience.

3.5.3 Solar Design Engineer: Proficiency in PVsyst / Helioscope, 3 years experience.

3.5.4 O&M Supervisor (per district): ITI Electrician / Diploma, 3 years experience.

3.6 CONSORTIUM

Maximum 2-firm consortium permitted. Lead member: minimum 60% stake.

Lead member must individually satisfy similar work experience criteria.

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SECTION 4: SCOPE OF WORK

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4.1 DESIGN AND ENGINEERING

4.1.1 Site survey and shadow analysis using drone/LiDAR for all 120 buildings

4.1.2 Structural load assessment and roof suitability report

4.1.3 Detailed system design using PVsyst (generation yield simulation)

4.1.4 Single line diagram, layout drawing, and as-built drawings

4.1.5 Net metering application and DISCOM liaison (JVVNL/JdVVNL/AVVNL)

4.2 SUPPLY AND INSTALLATION

4.2.1 MNRE-approved Monocrystalline Solar Panels (≥ 540 Wp each) – ~9,260 units

4.2.2 String/Central Inverters (grid-tied, BIS certified) – as per design

4.2.3 Module Mounting Structure (MMS) – hot-dip galvanised MS/GI, wind 150 kmph

4.2.4 AC/DC cables, junction boxes, surge protection devices (SPDs)

4.2.5 Energy meters (bi-directional, smart DLMS/COSEM protocol)

4.2.6 Remote Monitoring System (RMS) with GPRS/4G, web portal + mobile app

4.2.7 Lightning Arrestor, earthing, fire safety equipment at each site

4.2.8 Signage and safety barriers at all sites

4.3 GRID CONNECTIVITY AND METERING

4.3.1 Net metering connection with respective DISCOM

4.3.2 Supply of bi-directional energy meter (DISCOM approved)

4.3.3 Protection relay coordination and DISCOM inspection compliance

4.4 TESTING AND COMMISSIONING

4.4.1 Pre-commissioning checks: IV curve tracing, IR thermography

4.4.2 Performance Ratio (PR) measurement (minimum 75% at commissioning)

4.4.3 String-level and system-level testing and documentation

4.4.4 Witness testing by REDC / third-party inspection agency (ERDA/CPRI)

4.5 OPERATION AND MAINTENANCE (7 YEARS)

4.5.1 Scheduled preventive maintenance: quarterly at each site (4 times/year)

4.5.2 Panel cleaning: minimum 2 times/month (dusty Rajasthan conditions)

4.5.3 Performance monitoring: daily RMS data review, monthly energy reports

4.5.4 Annual thermography and IV curve testing

4.5.5 DISCOM coordination for grid faults and net metering reconciliation

4.5.6 System uptime SLA: 97% (measured per site, per quarter)

4.5.7 Response time: within 24 hours for critical faults, 72 hours for others

4.5.8 Annual Energy Production Guarantee: 90% of PVsyst-simulated P50 yield

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SECTION 5: BILL OF QUANTITIES (BOQ)

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SCHEDULE A – SOLAR PV SUPPLY AND INSTALLATION

S.No	Description	Unit	Qty	Rate(Rs.)	Amount(Rs.)
A.1	Monocrystalline Solar Panel ≥ 540 Wp (MNRE listed)	No.	9260	14500	134270000
A.2	String Inverter 50 KW (grid-tied, IP65)	No.	100	95000	9500000
A.3	Module Mounting Structure (per KWp, GI)	KWp	5000	4200	21000000
A.4	DC Cable 6 sqmm (FR-LSH, UV resistant)	m	180000	42	7560000
A.5	AC Cable 35 sqmm (armoured)	m	12000	185	2220000
A.6	DC Junction Box (8-string, weatherproof)	No.	620	3200	1984000
A.7	Surge Protection Device (DC + AC, Type 2)	Set	240	2800	672000
A.8	Bi-directional Energy Meter (DLMS/COSEM, DISCOM)	No.	120	8500	1020000
A.9	Remote Monitoring System (per site, GPRS/4G)	No.	120	12000	1440000
A.10	Lightning Arrestor + Earthing (per site)	Set	120	6500	780000

SCHEDULE B – CIVIL AND STRUCTURAL

B.1	Structural assessment and roof waterproofing	Sqm	48000	850	40800000
B.2	Conduit & Cable Tray installation (per site avg)	LS	120	18000	2160000
B.3	Safety barrier and signage (per site)	Set	120	4500	540000
B.4	Fire extinguisher (ABC 5 kg, per site)	No.	120	1800	216000

SCHEDULE C – GRID CONNECTION AND METERING

C.1	Net metering application charges (per DISCOM)	No.	120	5000	600000
C.2	Protection panel and DISCOM coordination	No.	120	22000	2640000

SCHEDULE D – ENGINEERING AND COMMISSIONING

D.1	Drone/LiDAR site survey and shadow analysis	No.	120	8000	960000
D.2	PVsyst detailed design report (per site)	No.	120	4500	540000
D.3	ERDA/CPRI commissioning inspection	No.	120	6000	720000
D.4	Training of building maintenance staff (per site)	No.	120	2500	300000
D.5	As-built drawings and documentation set	No.	120	3000	360000

SCHEDULE E – OPERATION AND MAINTENANCE (7 YEARS)

E.1	O&M Year 1 (365-day comprehensive)	LS	1	2400000	2400000
E.2	O&M Year 2	LS	1	2520000	2520000
E.3	O&M Year 3	LS	1	2646000	2646000
E.4	O&M Year 4	LS	1	2778000	2778000
E.5	O&M Year 5	LS	1	2917000	2917000
E.6	O&M Year 6	LS	1	3063000	3063000
E.7	O&M Year 7	LS	1	3216000	3216000

TOTAL ESTIMATED COST (A+B+C+D+E): Rs. 28,75,00,000/-

NOTE: All rates inclusive of GST, transportation, insurance, and incidentals.

Panel and inverter supply must be from MNRE-approved list valid at time of supply.

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SECTION 6: COMMERCIAL TERMS AND CONDITIONS

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6.1 PAYMENT TERMS

Stage 1: 5% on signing of Contract Agreement

Stage 2: 50% on pro-rata basis against supply and delivery at respective sites
(against GRN, inspection report, and REDC site engineer's certification)

Stage 3: 25% on commissioning of each site cluster (per district milestone)

Stage 4: 10% on receipt of net metering approval from respective DISCOM

Stage 5: 5% on submission and approval of first year energy generation report

Stage 6: Remaining 5% retained as retention money; released after DLP expiry.

O&M Payment: Bi-annual (every 6 months) in arrears, subject to:

- SLA uptime compliance verification by REDC
- Submission of maintenance logs, cleaning records, and energy reports
- PR not below 72% for the period (else proportionate deduction)

6.2 LIQUIDATED DAMAGES

6.2.1 Delay in Commissioning: 0.5% per week delay, maximum 7.5% of contract value.

6.2.2 Energy Production Shortfall: If actual generation < 80% of guaranteed P50
yield in any year, contractor liable at Rs. 4.50/- per unit shortfall.

6.2.3 Uptime SLA Breach: Rs. 10,000/- per site per month below 97% uptime.

6.3 PERFORMANCE GUARANTEE

Bank Guarantee: 5% of contract value, valid for 7 years + 90 days post-commissioning.

6.4 PRICE VARIATION

No variation for supply and installation phase.

O&M escalation: 5% per annum from Year 2 (built into Schedule E).

6.5 INSURANCE

Contractor's All Risk (CAR) including panels, MMS for full replacement value.

Panel Performance Insurance: 25-year linear degradation warranty from OEM
(not more than 0.7% degradation per year; minimum 80% output at 25 years).

Third Party Liability: Minimum Rs. 1 Crore per occurrence per site.

6.6 TECHNOLOGY UPGRADE OBLIGATION

Contractor shall replace any component failing to meet PR standards with
equivalent or superior technology at no additional cost to REDC within the O&M period.

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SECTION 7: RISK ASSESSMENT

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7.1 HIGH RISK ITEMS

7.1.1 Roof Structural Risk: Several older government buildings may have
deteriorated roof structures. Structural strengthening, if required, is
in contractor's scope. No extra claims for structural remediation.

7.1.2 DISCOM Interconnection Risk: Net metering approvals may be delayed by
DISCOM due to grid capacity constraints. Delays beyond 90 days attributable
to DISCOM shall be treated as Force Majeure for payment milestone only.

7.1.3 Panel Degradation Risk: High ambient temperatures in Rajasthan (up to 48°C) may accelerate panel degradation. Contractor to select panels with verified low temperature coefficient ($P_{max} \leq -0.35\%/^{\circ}\text{C}$).

7.2 MEDIUM RISK ITEMS

7.2.1 Dust Soiling Loss: Rajasthan dust storms (andhi) may reduce PR below targets. Contractor to factor in higher cleaning frequency; no force majeure relief for dust-related generation shortfall.

7.2.2 Grid Curtailment Risk: DISCOM may curtail solar injection during low-load periods. Documented curtailment events excluded from energy guarantee calculation.

7.2.3 Vandalism / Theft Risk: Panel theft and vandalism reported in similar projects. Contractor responsible for site security and insurance during O&M.

7.3 LOW RISK ITEMS

7.3.1 Labour Availability: Skilled electricians and solar installers may be scarce in smaller cities. Contractor to plan workforce deployment.

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SECTION 8: EVALUATION CRITERIA

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8.1 TWO-BID SYSTEM (Technical + Financial)

8.2 TECHNICAL EVALUATION (Pass/Fail + Scoring)

Criterion	Max Marks

Similar Work Experience (MWp completed)	30
Financial Capacity	10
Key Personnel Qualifications	15
Technical Methodology & Work Program	20
Panel & Inverter Make (MNRE list ranking)	15
O&M Plan and Local Presence	10
TOTAL	100

Minimum Qualifying Score: 65 out of 100

8.3 FINANCIAL EVALUATION

Lowest Valid Financial Bid (L1) basis.

L1 bidder invited for negotiation.

LCOE (Levelised Cost of Energy) calculation may be used for comparison if bids are within 5% of each other.

8.4 PREFERENCE

5% price preference for MSME-registered bidders as per GoR policy.

Domestic content requirement: Panels and inverters must be from MNRE-approved manufacturers with manufacturing facilities in India.

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SECTION 9: GENERAL CONDITIONS OF CONTRACT

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9.1 GOVERNING LAW

Governed by laws of India. Jurisdiction: Courts at Jaipur, Rajasthan.

9.2 DISPUTE RESOLUTION

Step 1: Mutual negotiation (21 days).

Step 2: Arbitration under Arbitration and Conciliation Act, 1996.

Sole arbitrator appointed by REDC CMD. Venue: Jaipur.

9.3 FORCE MAJEURE

Natural calamity, earthquake, flood, lightning strike, declared war,

DISCOM grid curtailment (documented). Notice within 5 days of occurrence.

9.4 TERMINATION FOR CONVENIENCE

REDC may terminate on 45-day notice. Payment for completed and accepted work.

9.5 TERMINATION FOR CAUSE

Immediate termination if:

- (a) Contractor abandons work at more than 10 sites simultaneously
- (b) Fraudulent billing of O&M services
- (c) Use of non-MNRE-approved components without written waiver
- (d) Contractor becomes insolvent

9.6 SUBCONTRACTING

Up to 25% of contract value may be subcontracted to MSME firms with prior written approval. Contractor remains wholly responsible for

quality, safety, and warranty obligations.

9.7 CHANGE ORDERS

REDC may add up to 10 additional sites (equivalent MWp) at BOQ rates within 12 months of contract award. No separate mobilisation charges.

9.8 INTELLECTUAL PROPERTY

Monitoring portal custom-developed for this project, including dashboards, APIs, and data schemas, shall be owned by REDC upon final payment. Third-party SCADA/monitoring software retains vendor licensing terms.

9.9 CONFIDENTIALITY

All building-level energy consumption data, occupancy patterns, and DISCOM tariff structures are confidential. Obligation: 3 years post-contract.

9.10 ENVIRONMENTAL AND SOCIAL

Contractor shall comply with Rajasthan State Pollution Control Board requirements. Proper disposal of packaging waste and old electrical components via authorised e-waste recyclers. Monthly environmental compliance reports to REDC.

9.11 SAFETY

All works to comply with IS 13947, National Electrical Code (NEC), and MNRE safety guidelines. PPE mandatory for all workers at height. Any fatal accident to be reported to REDC within 2 hours.

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SECTION 10: DOCUMENTS REQUIRED WITH BID

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ENVELOPE 1 – EMD AND FEES

- (a) EMD Instrument
- (b) Tender Document Fee payment receipt
- (c) Mandatory site visit attendance certificate

ENVELOPE 2 – TECHNICAL BID

- (a) Form of Tender (signed by authorised signatory)
- (b) Certificate of Incorporation / Registration document
- (c) GST Certificate, PAN, MSME/Udyam Certificate (if applicable)
- (d) Audited Financial Statements – last 3 years
- (e) Banker's Solvency Certificate (not older than 3 months)
- (f) Completion Certificates for similar solar works (attested)
- (g) MNRE channel partner / system integrator certificate
- (h) Panel manufacturer and inverter supplier authorisation letters
- (i) PVsyst simulation report for any reference project
- (j) CVs and degree certificates of key personnel
- (k) Detailed technical methodology and project execution plan
- (l) List of existing O&M sites with contact references
- (m) Self-declaration of non-blacklisting / debarment
- (n) Power of Attorney for authorised signatory

ENVELOPE 3 – FINANCIAL BID

- (a) Completed BOQ (Schedules A to E) with unit rates

(b) Summary price sheet and covering letter with total bid value

All three envelopes to be sealed separately and placed inside outer envelope
superscribed: "Tender No. REDC/SOLAR/2025-26/041 – Due 25 Feb 2026"

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END OF TENDER DOCUMENT

Rajasthan Energy Development Corporation Ltd | Jaipur

Tender Ref: REDC/SOLAR/2025-26/041

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